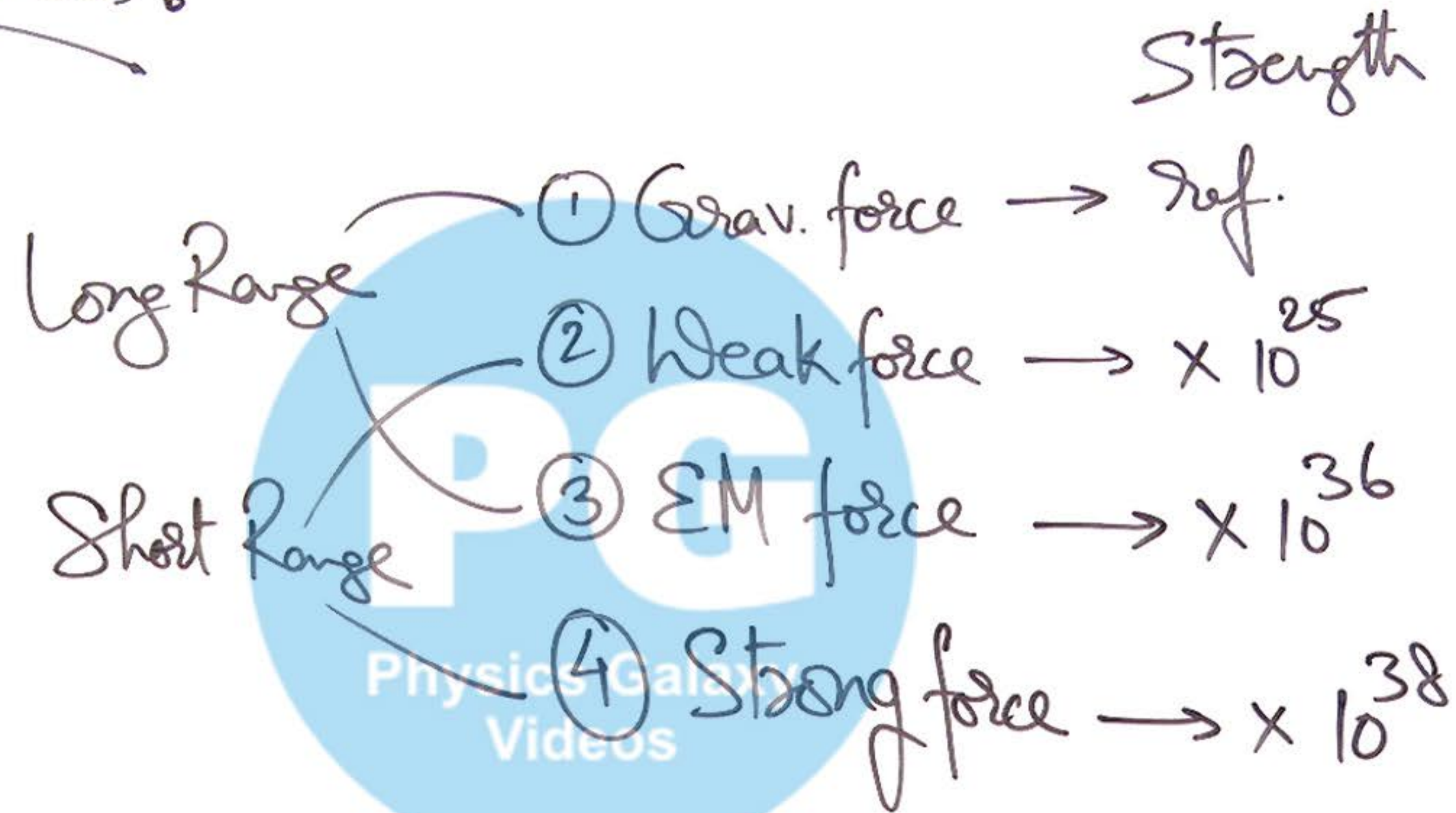


① Classification of Forces:

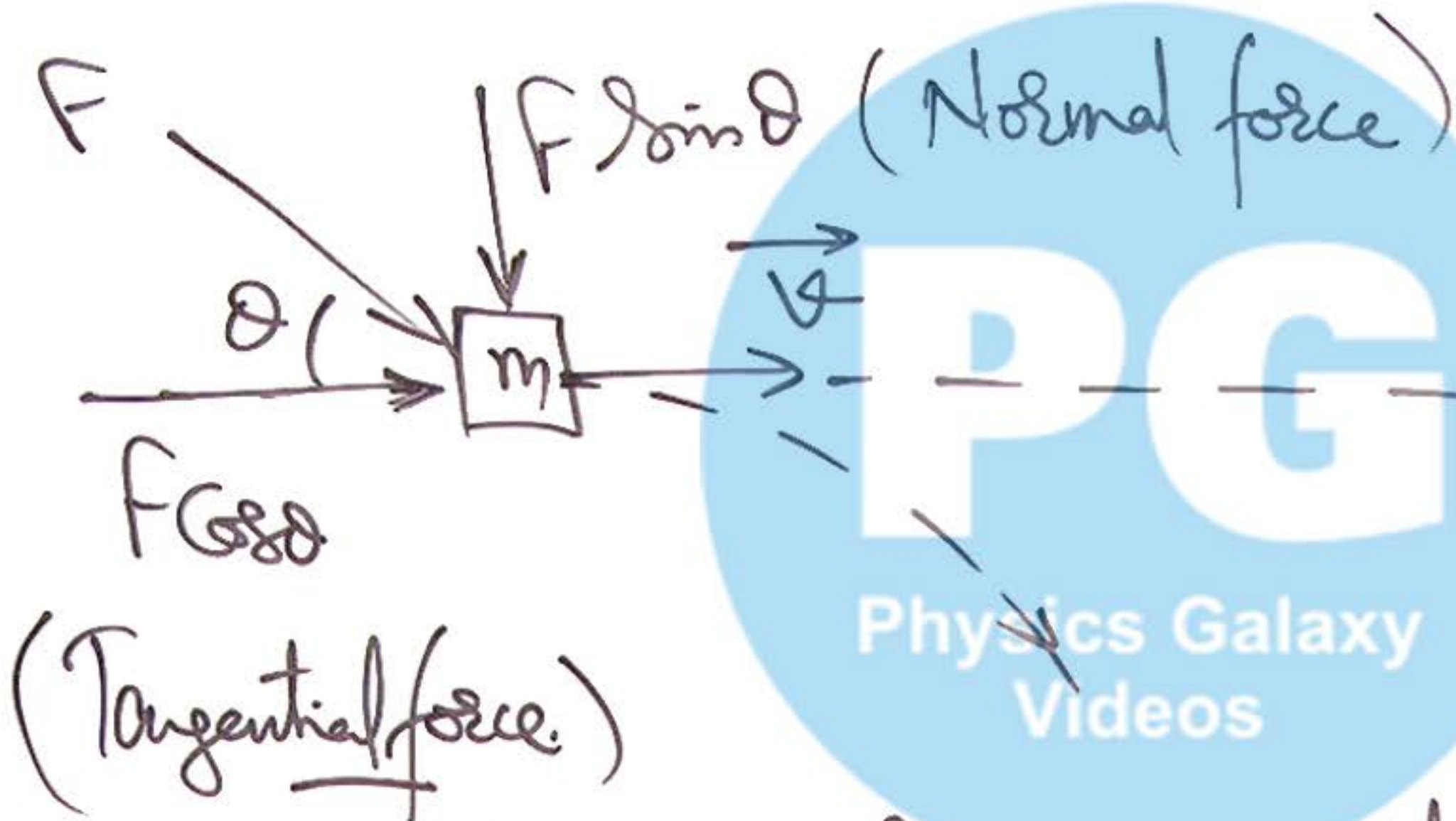


→ charges

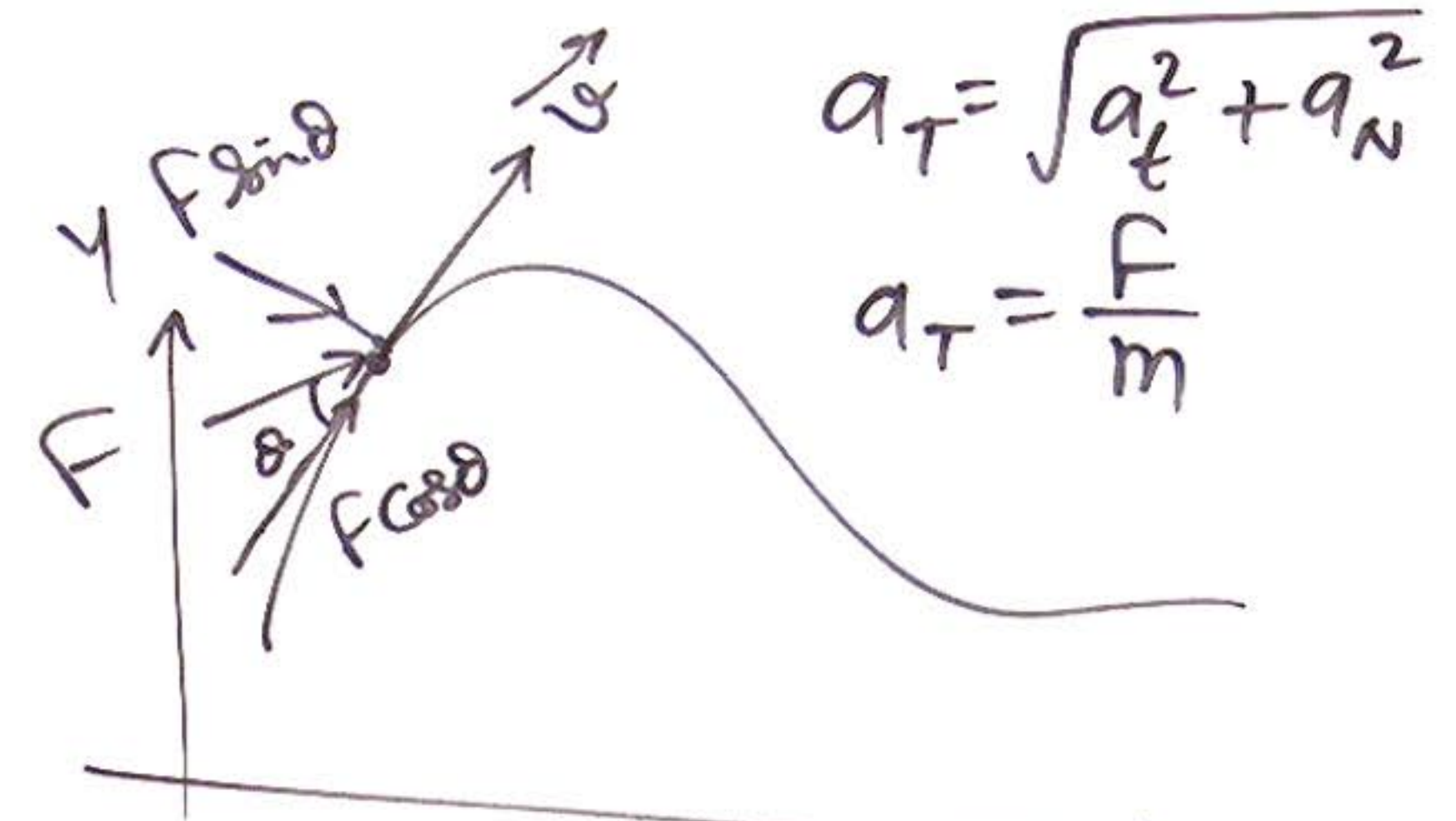
tion: $\vec{V}$

magnitude

direction



responsible for KE, Power, momentum mag. ---



$$a_T = \sqrt{a_t^2 + a_N^2}$$
$$a_T = \frac{F}{m}$$

$$- \left[\underset{\substack{\uparrow \\ \text{total acc}^n}}{\vec{a}_T} = \frac{d\vec{v}}{dt} \right]$$

Tangential accⁿ

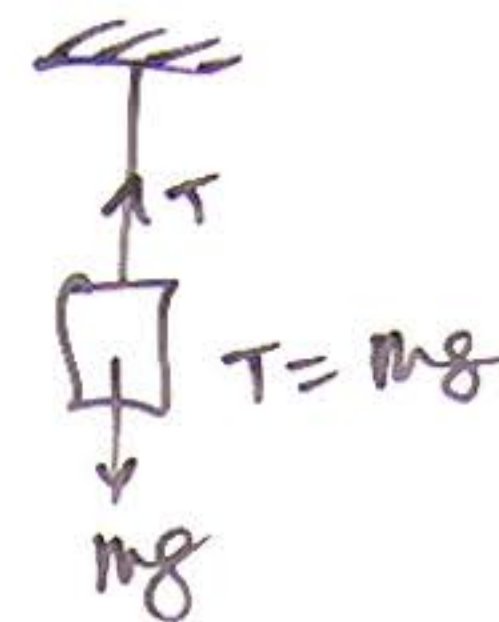
$$a_t = \frac{F \cos \theta}{m}$$

Normal accⁿ
 $\downarrow$ $a_N = \frac{F \sin \theta}{m} = \frac{v^2}{R}$
 Centripetal accⁿ

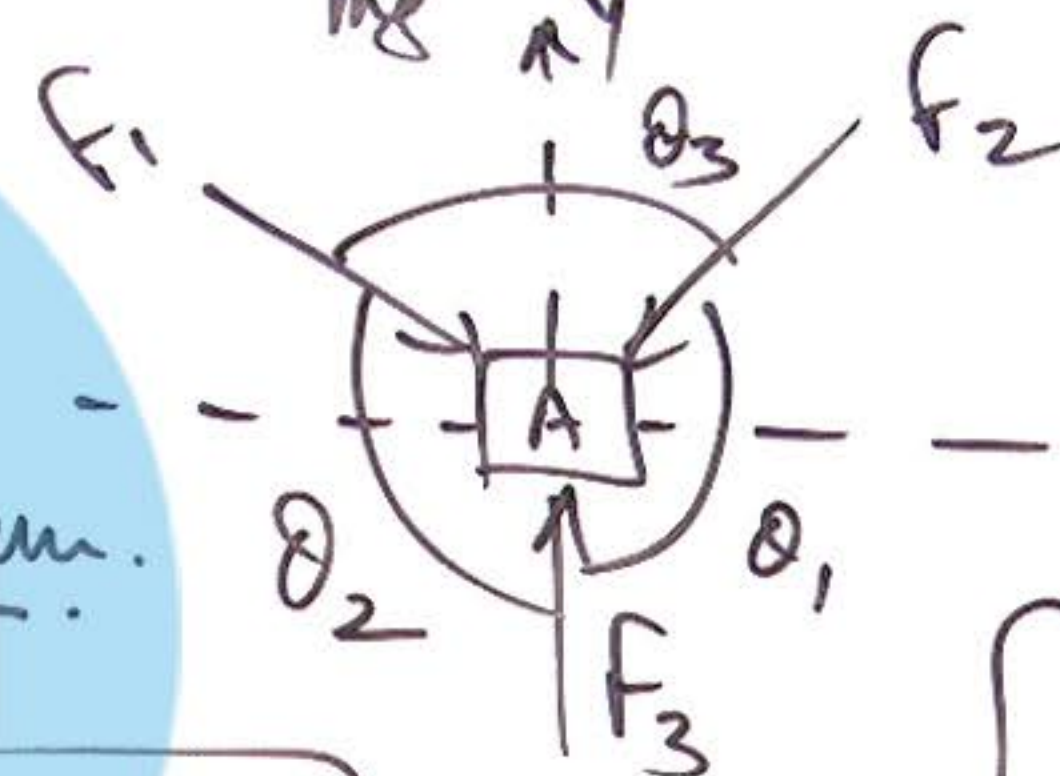
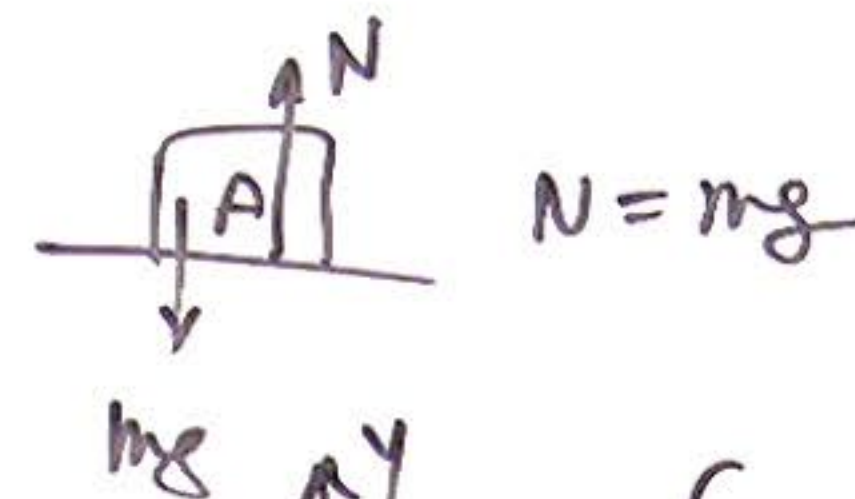
③ Static Equilibrium:

$$\vec{a}_{\text{net}} = 0 \iff \sum \vec{F} = 0 \text{ (Balanced forces)}$$

$$\vec{v} = \text{const} \implies |\vec{v}| = \text{const} \text{ dir } \nearrow$$



$$F_1 \rightarrow [A] \leftarrow F_2 \quad F_1 = F_2$$



Lami's Theorem.

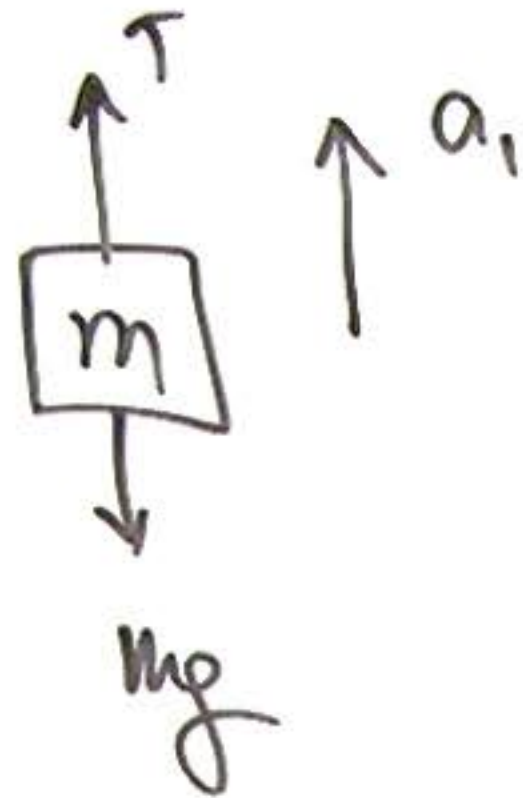
$$\frac{F_1}{\sin \theta_1} = \frac{F_2}{\sin \theta_2} = \frac{F_3}{\sin \theta_3}$$

$$\begin{cases} \sum F_x = 0 \\ \sum F_y = 0 \end{cases}$$

3 or more forces.

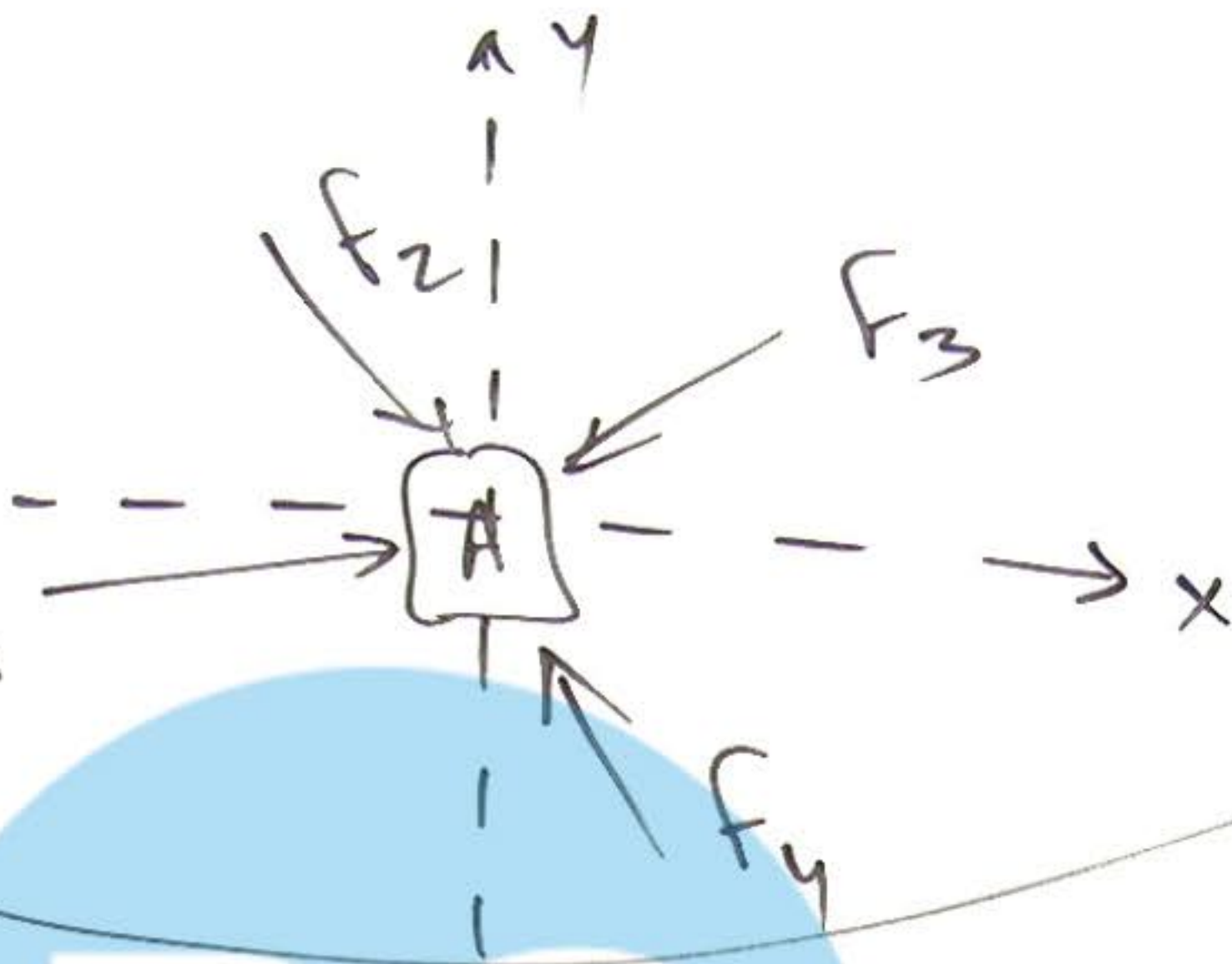
Unbalanced forces:

NL-II $\left[\sum \vec{F} = m \vec{a}_{\text{net}} \right]$



$$T - mg = ma_1$$

$a_1 \leftarrow$



NL-2 for SOP

$$\sum \vec{F} = \sum m \cdot \vec{a}_{\text{com}}$$

✓ $\left[\begin{aligned} \sum F_x &= m a_x \\ \sum F_y &= m a_y \end{aligned} \right]$

$$a_{\text{net}} = \sqrt{a_x^2 + a_y^2}$$



$$a_x = a_1 - a_2 \cos \theta$$

$$a_y = a_2 \sin \theta$$

⑤ Action-Reaction of Common forces:

External $\rightarrow$ F, mg

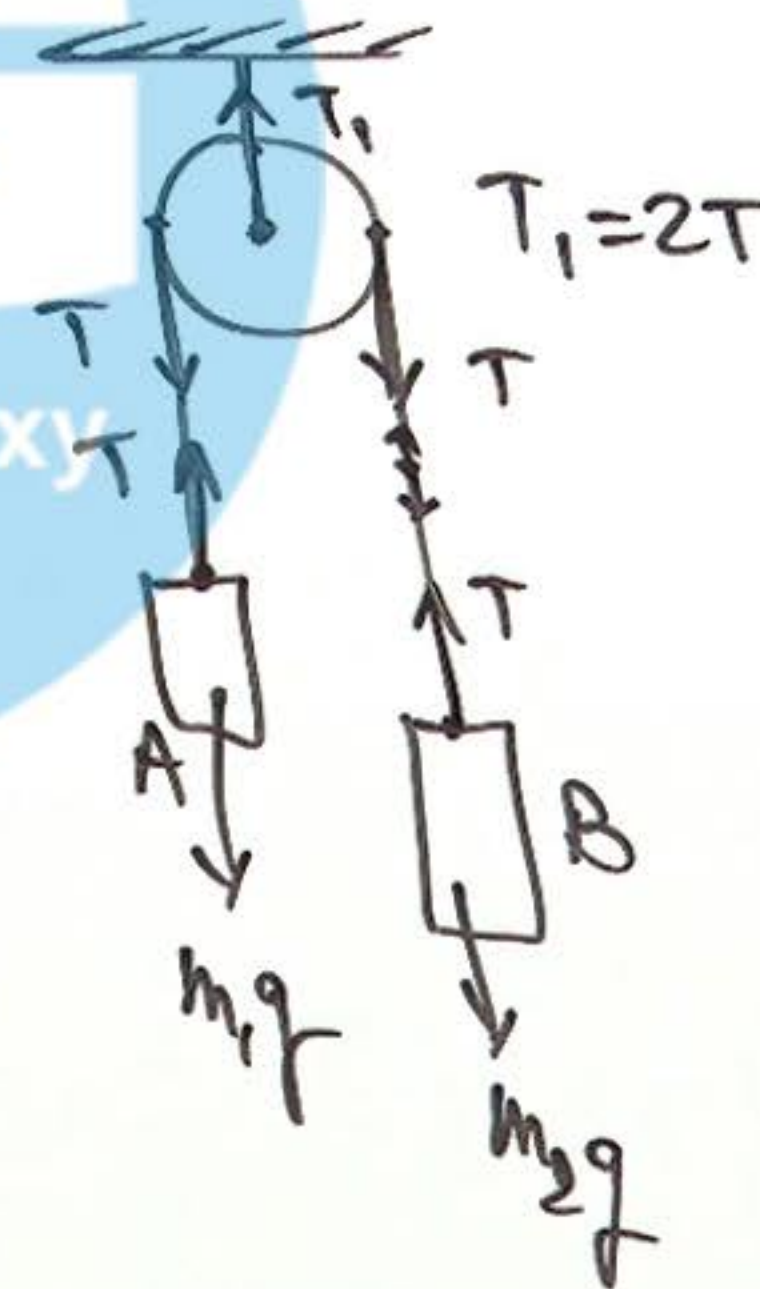
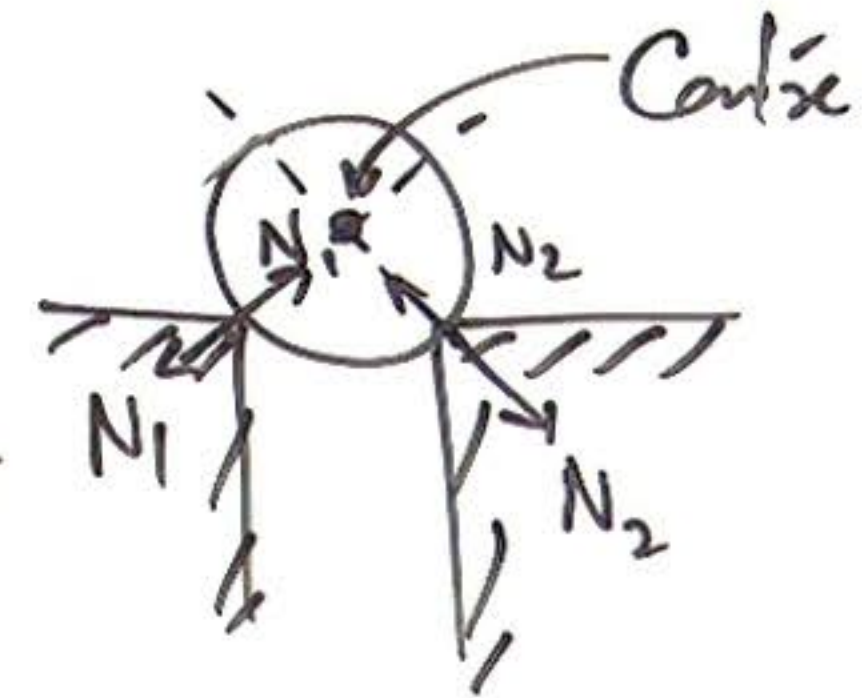
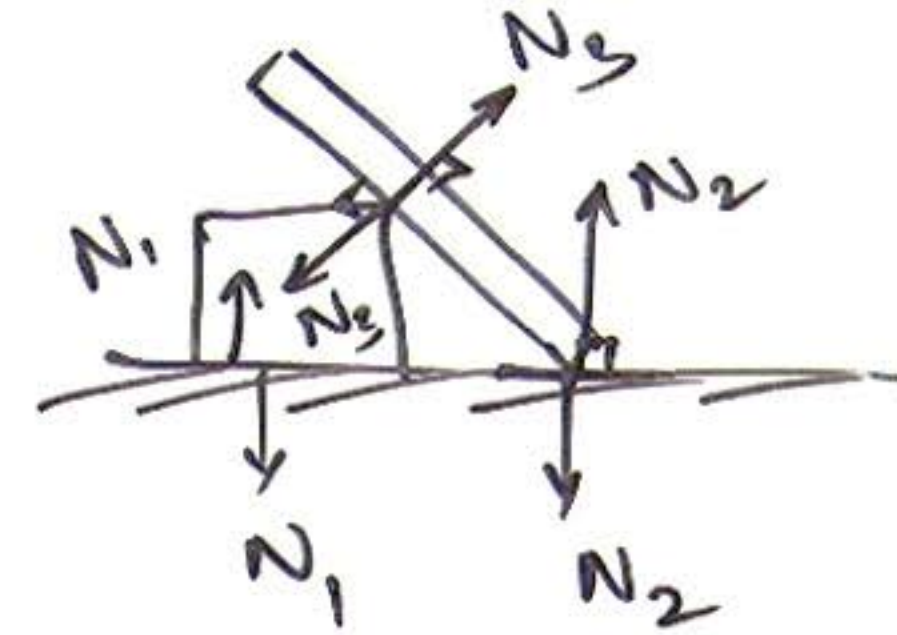
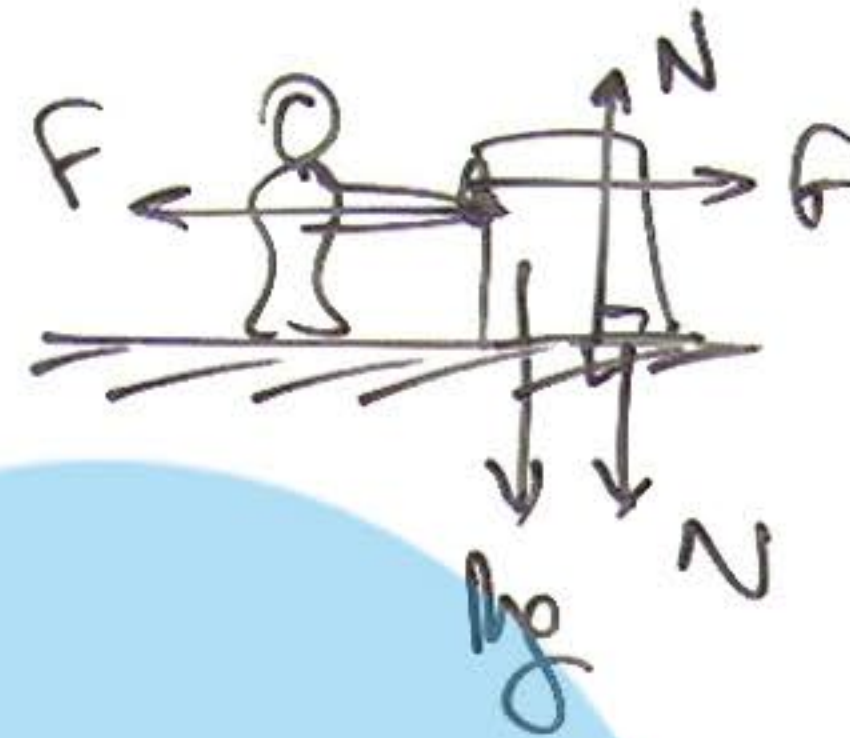
Internal $\rightarrow$ $T, N \checkmark$ & f

$$\vec{\Sigma F} = 0$$

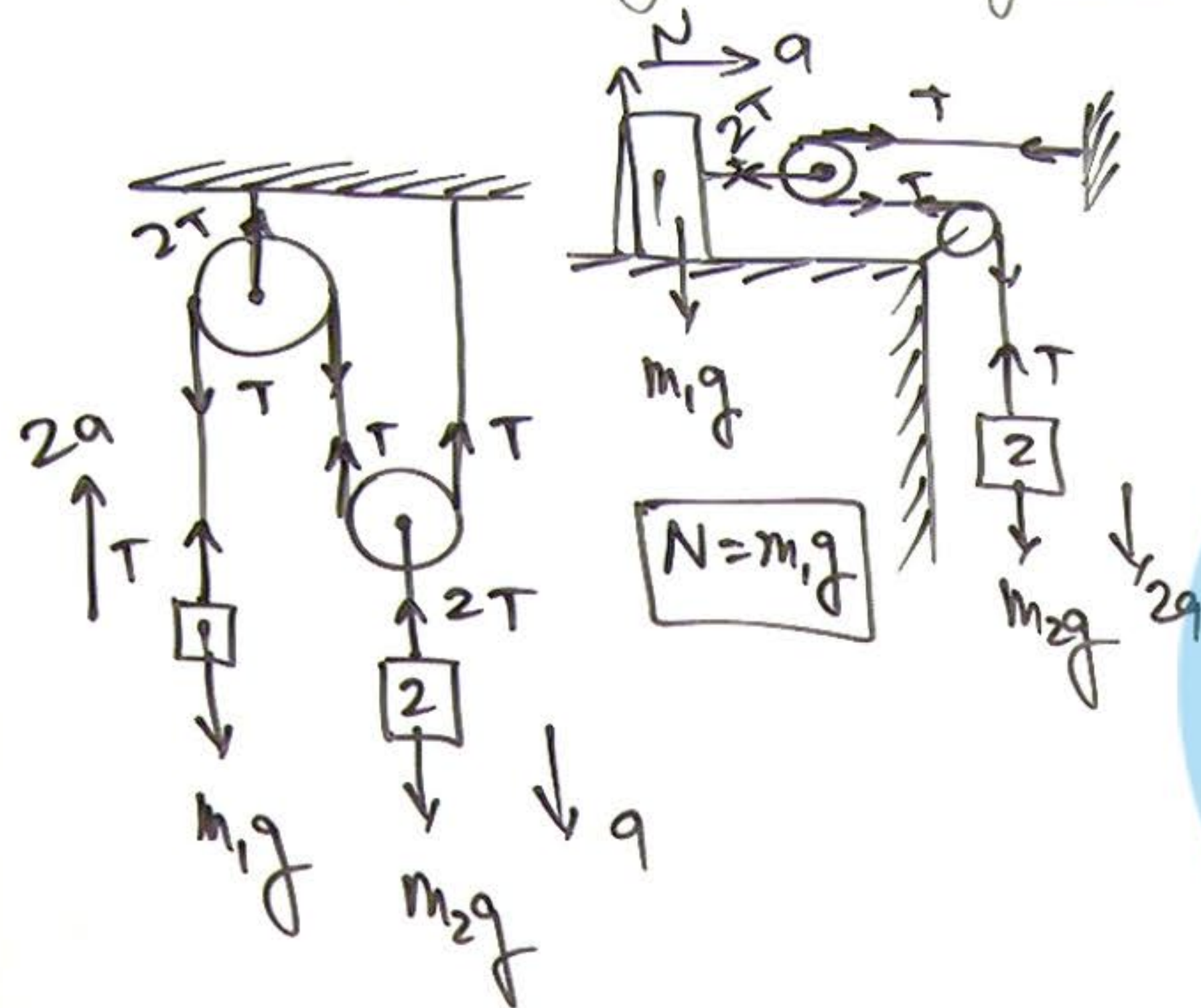
for all massless bodies.

Normal Contact force.

It Acts $\perp$ to Contact Surfaces



⑥ Appⁿ of Pulley-Block System :

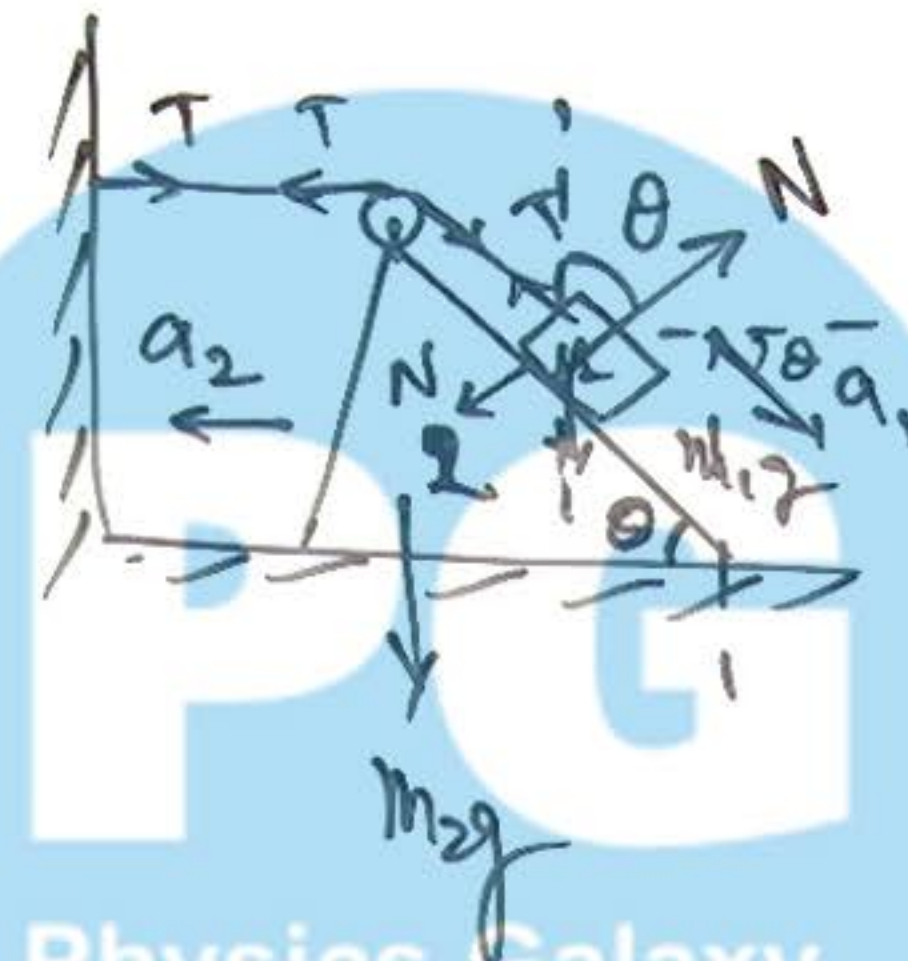


$$m_2g - 2T = m_2a$$

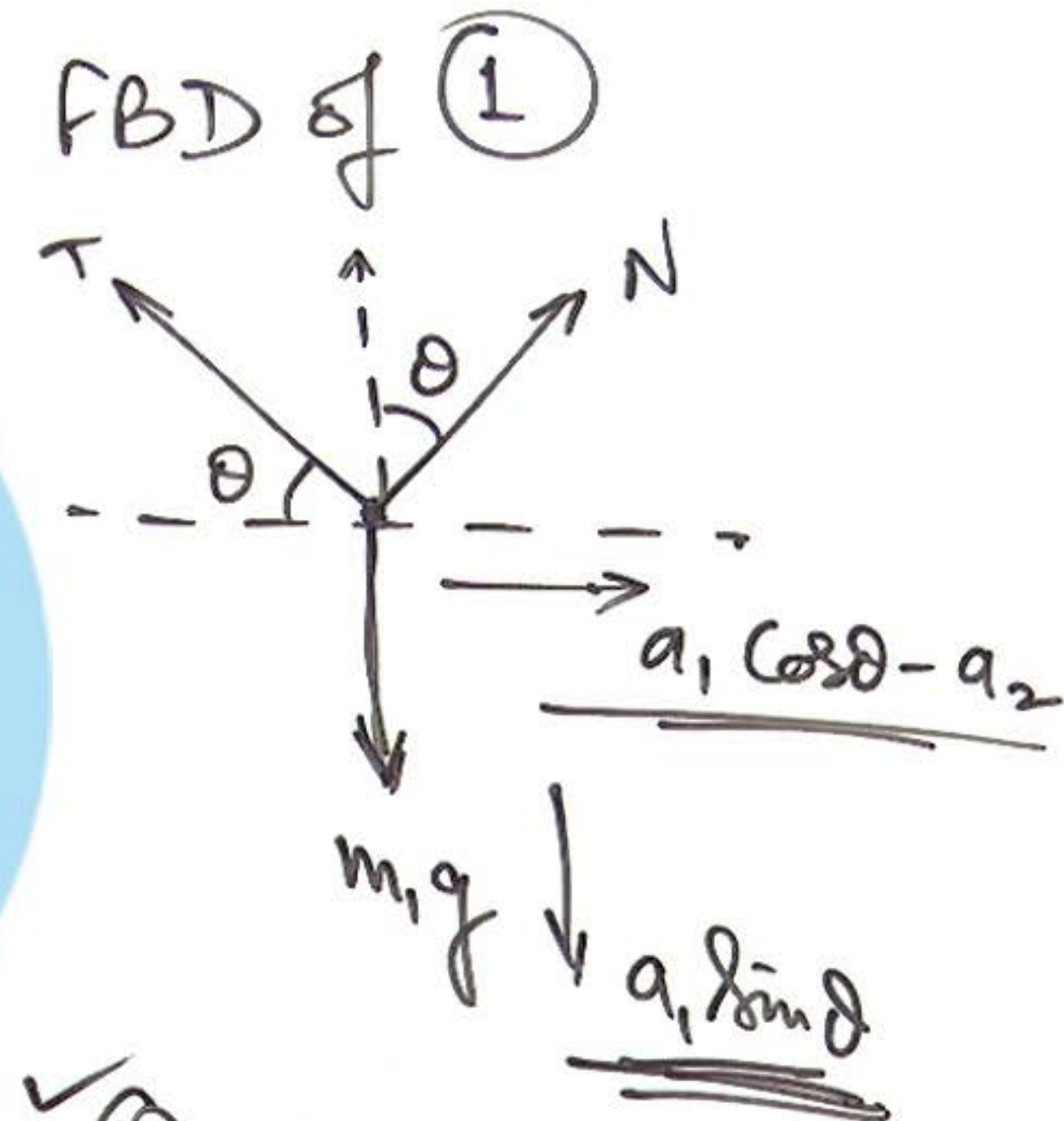
$$T - m_1g = m_1(2a)$$

④ $\rightarrow \Sigma F_y = ma_y$

$$m_1g - N \cos \theta - T \sin \theta = m a_1 \sin \theta$$



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⑤ $\rightarrow \Sigma F_x = ma_x$

$$N \sin \theta - T \cos \theta = m (a_1 \cos \theta - a_2)$$

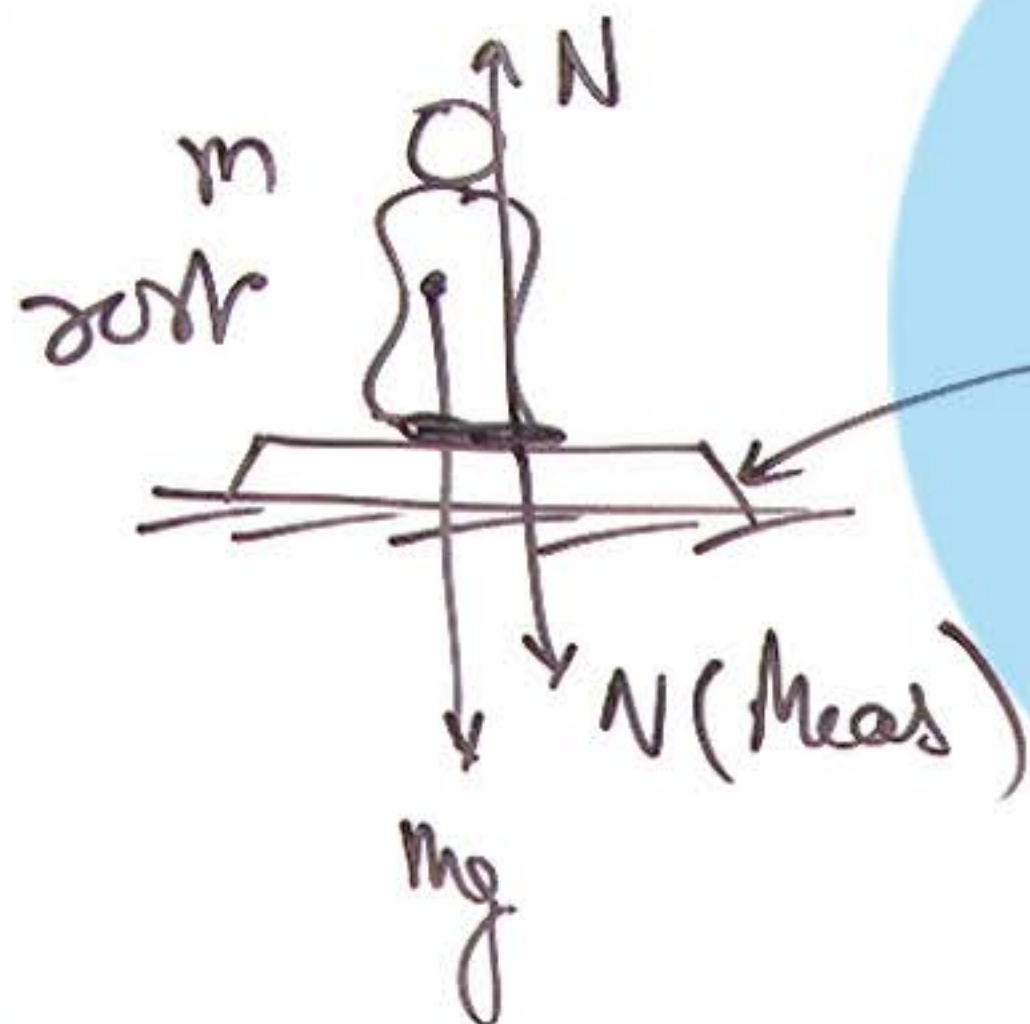
⑦ Concept of Weighing Machine:

Measures Normal reaction
on surface of M/c.

FBD of man.



$$N = mg$$



$$\text{Reading} = \frac{N}{g} = m \text{ kg.}$$



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for (m_1)

$$\text{Reading} = \frac{N}{g} \text{ kg}$$

$$N - m_1 g = m_1 a$$

$$N = m_1 (a + g)$$

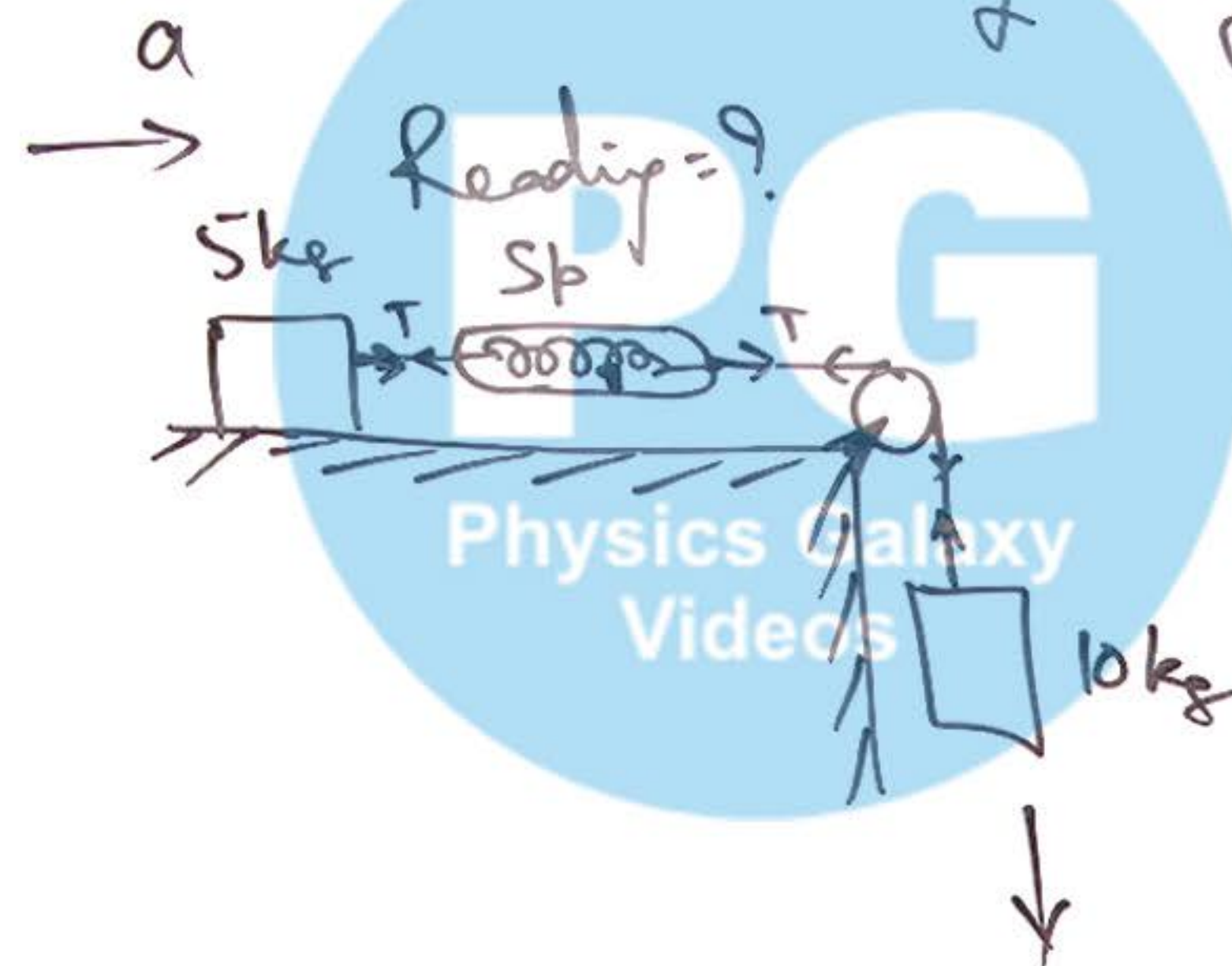
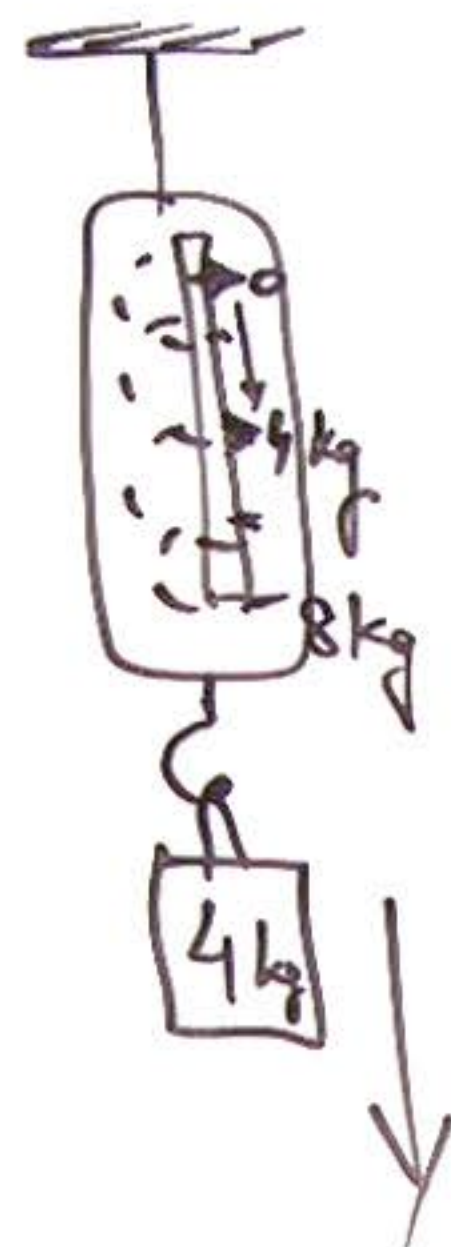
$$a = \left[\frac{M - (m + m_1)}{M + (m + m_1)} \right] g$$

⑨ Cases of Spring Balance:

Spring force $F = kx$

It measures tension in string attached to spring in kgf.

$$\text{Reading} = \frac{T}{g} \text{ kg}$$



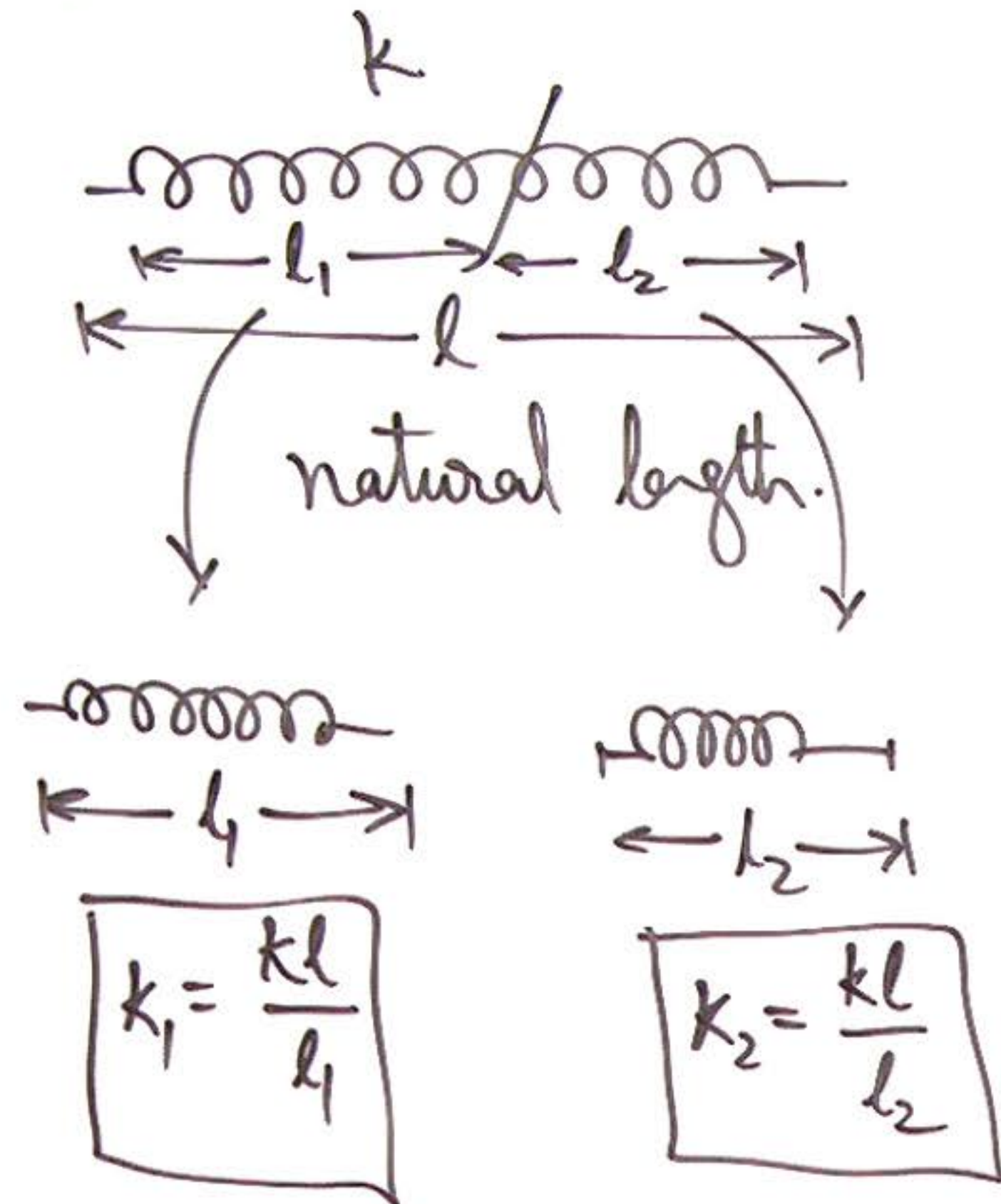
$$a = \frac{10g}{10+5} = \frac{100}{15} = \frac{20}{3} \text{ m/s}^2$$

tension in string

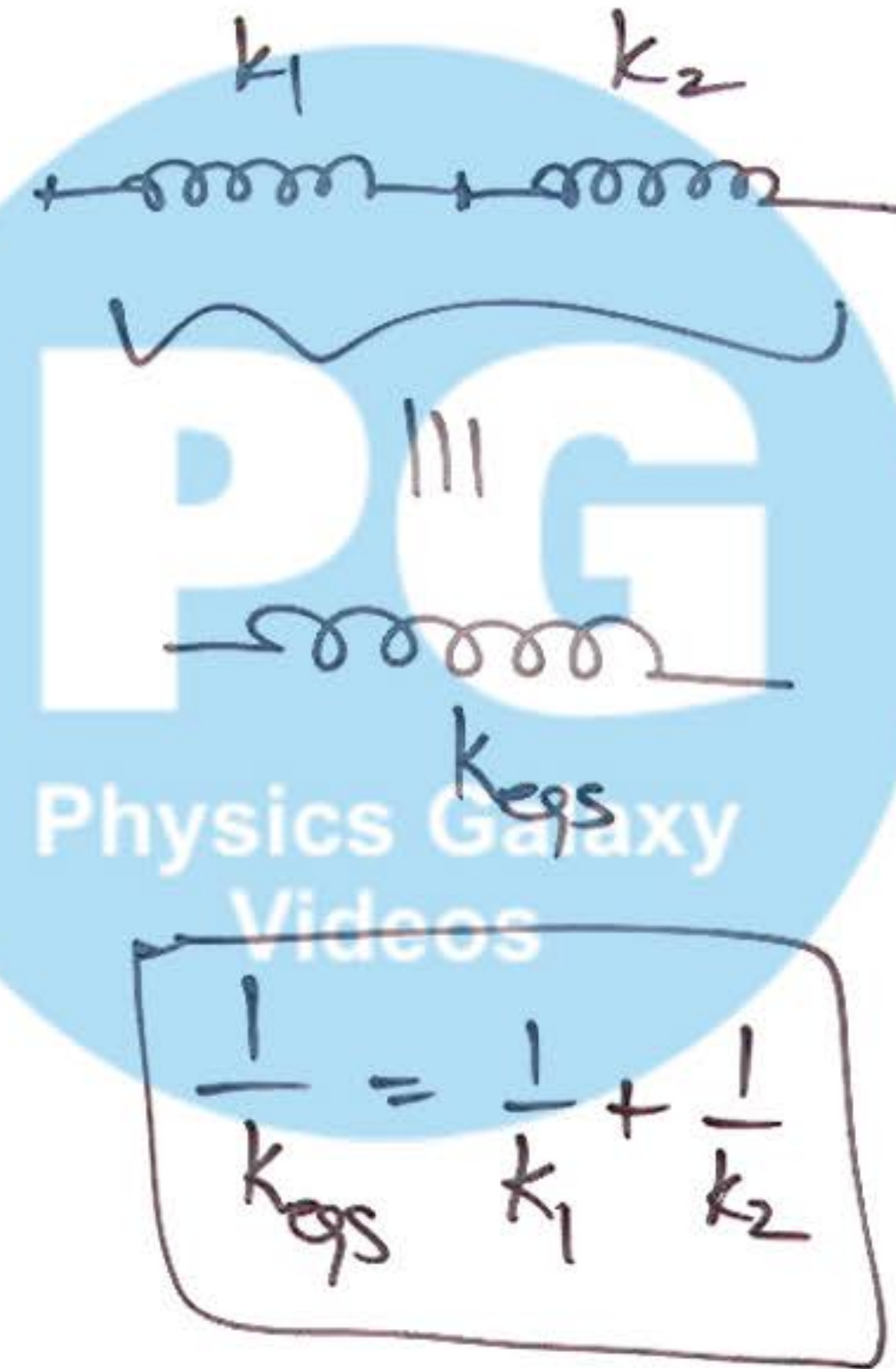
$$T = 5a = 5 \times \frac{20}{3} = \frac{100}{3} \text{ N}$$

$$\text{Reading of Sp} = \frac{T}{g} = \frac{10}{3} \text{ kg} = 3.33 \text{ kg}$$

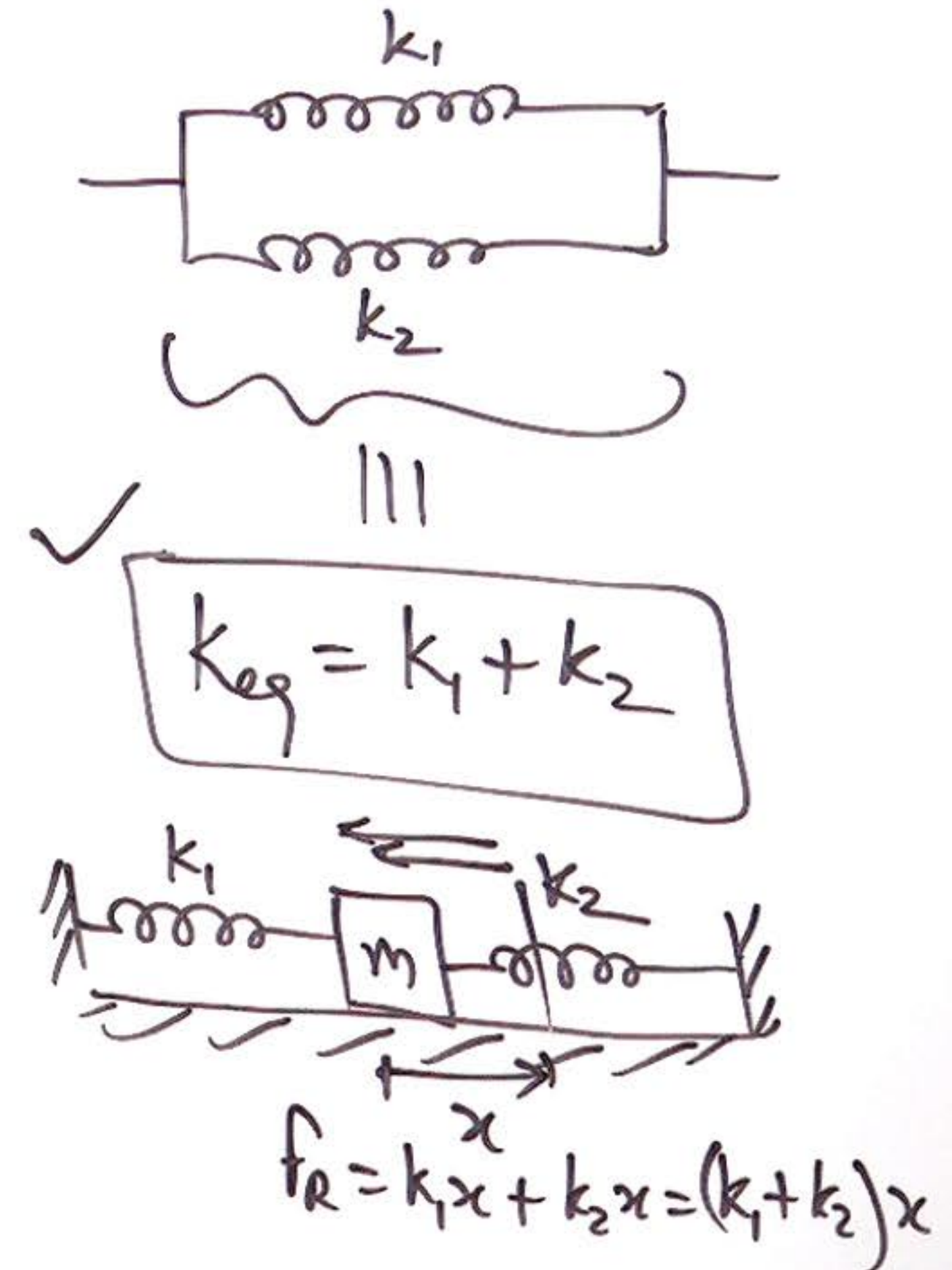
⑨ Cutting & Combination of Springs :



Series Combination



Parallel Combination



⑩ Elevator Cases:

Diagram of a person in an elevator accelerating downwards with acceleration a_1 . The person has mass m . The forces acting on them are Normal force N (up) and weight mg (down). The effective weight is $W_{eff} = m(g - a_1)$.

Reading = $\frac{N}{g} = \frac{m(g - a_1)}{g}$ App

for boy $N = m(g - a_1)$

net accⁿ of blocks

$a_{m_1} = (a_1 + a) \uparrow$

$a_{m_2} = (a_1 - a) \downarrow$

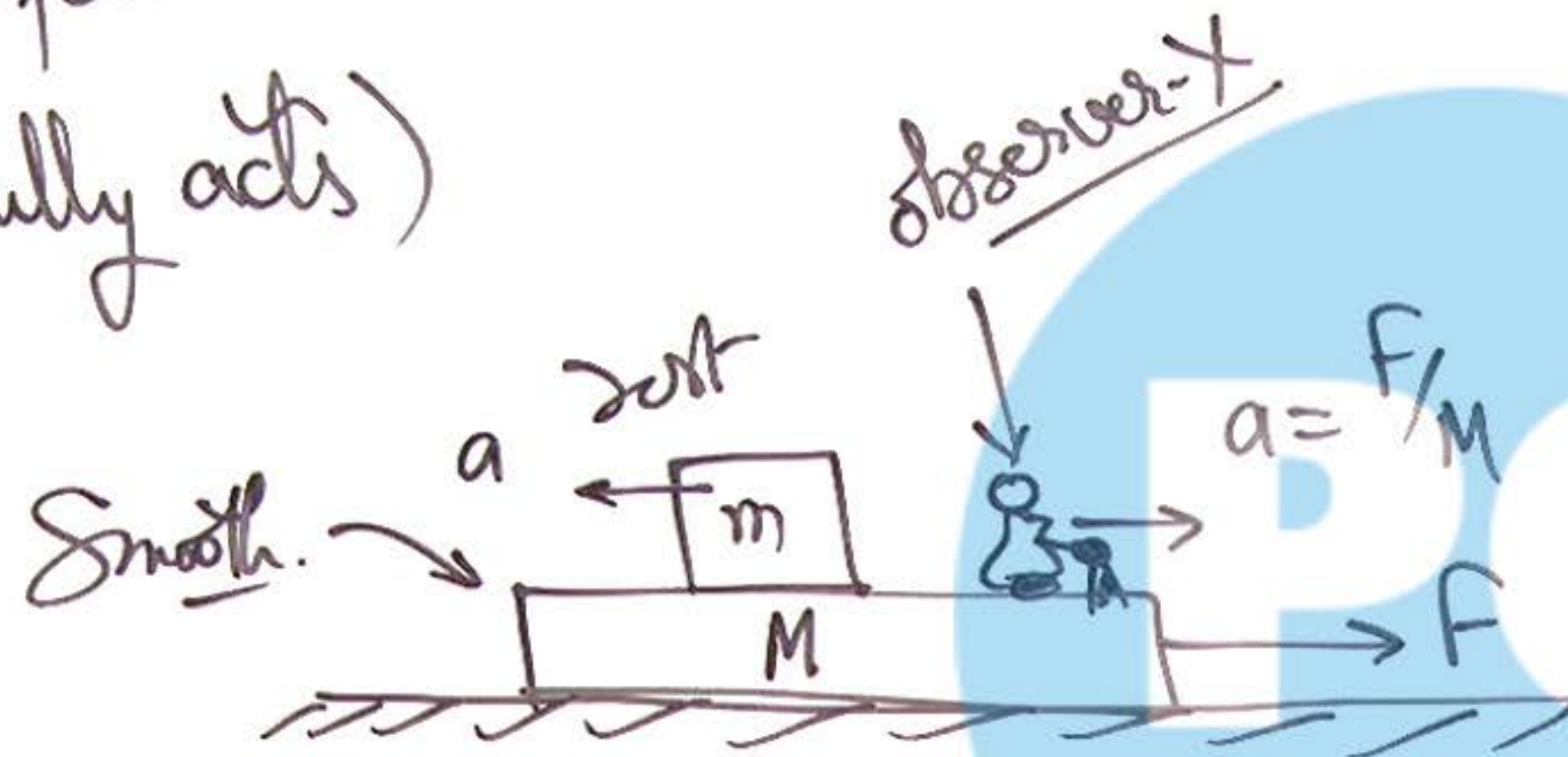
Diagram of an Atwood machine inside an elevator accelerating upwards with acceleration a . The elevator is labeled "observer". The masses are m_1 and m_2 , with $m_2 > m_1$. The acceleration of the blocks relative to the elevator is a_1 . The effective acceleration of the blocks is $a_1 = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) (g + a)$.

relative gravity / effective gravity
w.r.to elevator $g_{eff} = g + a$

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⑪ Pseud Force:

↓
false force
(not actually acts)



force observed by obs-x

on 'm' is

Pseudo force.

$$F_p = ma \text{ (leftward)}$$

If observer $acc^n = 0$

⇒ Ref frame is Inertial

If observer $acc^n \neq 0$

⇒ Ref frame is Non-Inertial